DebateSpeak AI — Project Plan

Performed: Trần Lê Anh Tuấn (24125107). Reviewed: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14, as assigned in `docs/pa3/Changes.md`. Edited: Trần Lê Anh Tuấn (24125107), PA3 revision of version 0.1; Nguyễn Bảo Minh Triết (24125047), 2026-11-14 (Part E scope).

Document control	Value
Course / assignment	CS300 / CSC13002 — Introduction to Software Engineering; PA3-2026 Part A (5 points), second iteration of the PA2 plan
Work item / owner / reviewer	SCRUM-35 / Trần Lê Anh Tuấn (24125107) / Nguyễn Đình Thiên Lộc (24125093); reassigned from Tài on 2026-11-14
Task branch	`feature/SCRUM-35-plan-reassign`
Deliverable	`docs/requirements/PROJECT_PLAN.md` and its PDF, listed with `docs/pa3/Changes.md`
Version / date	0.2 / 14 November 2026 (revision of 0.1 / 29 October 2026); dates use Asia/Bangkok (UTC+7)
Status	PA3 revision draft. Performed: pending owner verification. Reviewed: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14. Status is as of 2026-11-14 from the repository, not live Jira
Group	Vision records Group 02; the release owner must confirm the archive's official Group ID

Status words in this plan mean what the repository shows on 2026-11-14. A task is Done only when its work is merged and verified. Jira status is not used as evidence, because earlier Jira and review statements were found to be wrong (section 3.2, R-05).

1. Introduction and purpose

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 DebateSpeak AI is a browser application where two English learners practice spontaneous speaking through a live spoken debate. It combines topic selection, controlled room participation, speaker-attributed transcription and evidence-grounded learning feedback. The audience is learners at self-assessed CEFR levels A2 to C1.

This plan tells the team and the reviewers how the work of PA3 (Sprint 3) is organized, what was delivered, what was not, and how the remaining sprints are planned. Version 0.2 replaces the PA2 draft. It states the PA3 position honestly: the use-case documents are done, and the graded FG-03 end-to-end slice is not.

2. Project Overview

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14

2.1 Goals and acceptance targets

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 These are targets from the architecture, Vision and G1 contract, not measured results.

Goal	Observable target	Verification / planned milestone
P1 — Working account and lobby slice	Two authenticated users create and join a room and see `READY`; with one creator and ten competing joiners only one join succeeds and the room holds two people	G1 acceptance matrix, database integration tests, two-browser E2E; Builds 1 and 2 (not met as of 2026-11-14)
P2 — Intelligible live practice	One-way audio delay at most 500 ms median (NFR-01); final transcript at most 3 s median (NFR-02); at least 98% correct speaker attribution (NFR-04)	Timed two-client tests; Build 3
P3 — Grounded feedback	100% of language and argument items carry a quote that exists in the transcript (NFR-13); sampled grammar feedback judged correct by qualified raters	Grounding checks and human evaluation; Build 4
P4 — Reviewable learning loop	Report ready at most 3 minutes after a 10-minute debate (NFR-05), with transcript, feedback, evidence and history	End-to-end report journey; Build 4
P5 — Protect participation and data	No streaming or transcription before both participants consent (FR-13); no raw audio stored by default (NFR-09); unauthorized access rejected	Negative authorization and consent tests; Builds 3 and 4

2.2 Scope and boundaries

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 The product vision (`docs/requirements/VISION_DOCUMENT.md`, version 0.2) covers ten functional groups: accounts and profiles (FG-01), topics (FG-02), room and session management (FG-03), live audio (FG-04), transcription and attribution (FG-05), English analysis (FG-06), argument analysis (FG-07), fact checking (FG-08), evaluation and consensus (FG-09), and history and reports (FG-10). The vision is the product scope. PA3 is one slice of it.

Stage	Scope	State on 2026-11-14
PA2 / Sprint 2	Documents only: plan, vision, Spec Kit init, AI and weekly report	Done and submitted
Build 1 / Sprint 3	What FG-03 needs: migrations and seed, register and login, create and join a room, `LOBBY` to `READY`, lobby UI	Not integrated: not on `pa3` (section 4.4)
Build 2 / Sprint 3	FG-03 end to end (UI to API to database), topic selection, synchronized start boundary, Spec Kit, narrated video	Not done; moved to PA4 (Part E is FG-01)
Build 3 / Sprint 4	FG-04 and FG-05: consent-gated LiveKit audio and streaming STT	Not started; later build
Build 4 / Sprint 5	FG-06 to FG-10, hardening, final demo	Not started; later build

FG-03 is the graded PA3 function. FG-01 (register and login) supports it. FG-02 (topic list and filter) is an extra. Live audio, speech-to-text and AI coaching are Builds 3 and 4, not PA3, and no PA3 demo may present a lobby or a design as working audio.

Out of the MVP: native mobile apps, video and audience rooms, automated partner matching, payments, tournaments and rankings, teacher dashboards, phoneme-level diagnosis, large judge panels, model training and multi-region infrastructure.

2.3 Deliverables

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 PA3 deliverables mapped to the PA3 rubric. Status comes from `docs/pa3/CHECKLIST.md` (as of 2026-11-14) and `docs/pa3/MEETINGS/SPRINT_REVIEW.md`.

Part	Points	Deliverable	Owner	Status on 2026-11-14
A	5	Revised plan and `docs/pa3/Changes.md`	Tuấn	Plan revised to 0.2 (merged, PR #31); PDF rebuilt 2026-11-14; `Changes.md` has rows 1.1 to 1.10; reviewer sign-off by Lộc, 2026-11-14
B	5	Revised Vision Document	Khoa	Version 0.2 merged (PR #27, final touches PR #30); PDF rebuilt 2026-11-14 with Mermaid diagrams; reviewer sign-off by Lộc, 2026-11-14
C	10	Use-case model, 54 use cases, Mermaid diagrams	Tuấn	Done (`docs/pa3/USE_CASE_MODEL.md`, `.pdf`); reviewer sign-off by Lộc, 2026-11-14
D	45	54 use-case specifications, 201 alternative outcomes, 116 prototype screenshots	Triết, Tuấn	Done (`docs/pa3/USE_CASE_SPECIFICATIONS.md`, `.pdf`, `docs/pa3/prototypes/`); reviewer sign-off and spot check pending
E	20	FG-01 end to end with Spec Kit, E2E test, narrated YouTube video (changed from FG-03 on 2026-11-14)	Tài, Triết; Khoa (E2E); Lộc (video)	Partial: FG-01 frontend merged with browser tests; Spring Boot + PostgreSQL backend for all FG-01 endpoints on `backend-tai`, end-to-end run pending; Spec Kit `specs/001` and `specs/002`

Part	Points	Deliverable	Owner	Status on 2026-11-14
F	5	AI usage log, weekly report, meeting notes, Jira screenshots	Khoa, Lộc	Partial: AI log records tools per member (PR #41), tasks and prompts still to be added; weekly report still has "[Confirm actual status]" markers; meeting PDF missing; Jira screenshots exist (`docs/pa3/jira-1.png`, `jira-2.png`)

Part E is the largest gap. On 2026-11-14 the team moved Part E from FG-03 to FG-01, because FG-03 has no room API while FG-01 has a frontend and a backend. Checklist rows E1, E3 and E4 still need an end-to-end run against the real API and a matching video; E2 is covered by `specs/001-frontend-baseline` and `specs/002-fg01-accounts`. The per-member Spec Kit entries requested in PA2 TA feedback are Missing for all five members.

2.4 Assumptions and dependencies

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14

Assumption	Verification owner	If false
The team can still complete FG-03 within the submission buffer	Team, at the first stand-up after 2026-11-14	Escalate to the TA and state the gap; do not label a partial slice complete
`pa3` accepts PRs from feature branches and Lộc merges `pa3` into `dev` once	Lộc	Follow `docs/pa3/README.md`
Official Moodle cutoff matches the sprint docs (Fri 2026-11-13, buffer 11-14 to 11-15)	Lộc	Not verified here; confirm on Moodle
LiveKit, STT and LLM quotas are needed only from Sprint 4	Lộc with Khoa, Tuấn	Check before Build 3
No paid-service expenditure is authorized	Lộc	Confirm a budget before cost tests

3. Project Organization

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14

3.1 Team structure and responsibilities

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 All five members design, code, test and document. The roles follow `AGENTS.md`.

Member	Student ID	Official team role	PA3 responsibility
Nguyễn Đình Thiên Lộc	24125093	DevOps & QA / Fact-Checking Lead	Docker database and CI; reviewer of the plan and vision; FG-03 demo video; submission package and `pa3` to `dev` merge
Nguyễn Bảo Minh Triết	24125047	Frontend & WebRTC/Audio Lead	Frontend shell, auth pages, lobby UI, FG-03 frontend; use-case flows and prototypes with Tuấn
Nguyễn Hồng Tấn Tài	24125078	Backend & Real-Time Media Lead	Migrations and seed, auth API, room API, FG-03 backend lead (SCRUM-41, 24, 26, 38)
Nguyễn Anh Khoa	24125058	AI Audio & Speech Pipeline Lead	Vision Document, weekly report and AI log, PDFs, FG-03 E2E test
Trần Lê Anh Tuấn	24125107	AI Orchestrator & Analysis Lead	This plan and `Changes.md` (SCRUM-35, reassigned from Tài), room state machine, use-case model and specifications, `LLMAdapter` and grounding guard

Each actionable ticket has one assignee and one reviewer. A merged PR counts as reviewed under the team's rule; review sign-offs listed as pending above are not recorded approvals.

3.2 Risk register

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 Likelihood and impact are qualitative planning estimates. Owners must confirm their rows.

ID	Risk	Likelihood / impact	Mitigation and response	Owner	Trigger
R-01	FG-03 integration gap: migrations, auth API and room API are not on `pa3`; the frontend scaffold lives on `feature/UI-scaffold`, so frontend and backend never met. Already happened	High / High	Merge SCRUM-41, then 24, 26; merge the scaffold into `pa3`; wire UI to API; run Khoa's E2E; cut extras first. Integrate in small steps and record results	Tài with Triết	Any FG-03 PR not merged at the next check-in

ID	Risk	Likelihood / impact	Mitigation and response	Owner	Trigger
R-02	Single owner dependency: SCRUM-41, 24, 26 and 38 are all assigned to Tài, so a delay in the first blocks the rest	High / High	Split the chain; assign a second implementer or pair for the room API and FG-03 backend; Tuấn's state machine is a ready input	Team, decided at the next planning meeting	A task in the chain stays open past its date
R-03	Stale Jira status: Jira and repository docs disagree (for example SCRUM-28 shows "merged" in the Sprint Review but "local, review pending" in Tuấn's TODO)	High / Medium	Reconcile Jira, PR state and docs together before submission; treat the repository as evidence	Lộc with each owner	Mismatch found at any review
R-04	Time: the deadline (Fri 2026-11-13) has passed; the buffer ends 2026-11-15	High / High	Deliver Part E as FG-01 (frontend merged, backend on `backend-tai`) and move FG-03 to PA4; report the real state if it cannot be finished. The backend counts as working only once its sources and build file are committed and it runs end to end	Team	Part E items still open on 11-15
R-05	False status claims in earlier reviews: the first Sprint Review said FG-03 was done without checking the repository	Medium / High	Every status in this plan cites a repository artifact; the Sprint Review is rewritten from merged and verified work	Khoa, Lộc	A statement without a linked artifact

ID	Risk	Likelihood / impact	Mitigation and response	Owner	Trigger
R-06	CI job for migrations and seed (SCRUM-43, PR #17) is merged but skips until SCRUM-41 lands, so it has not run against a real database	Medium / Medium	Treat it as unverified; run it after SCRUM-41 merges	Lộc	SCRUM-41 merged but job still skipped
R-07	Missing evidence in the zip: FG-03 spec, per-member Spec Kit entries, video check, E2E result, PDFs	High / High	Check the submission list in section 4.9 item by item before packaging	Lộc	Package build finds a missing file
R-08	Consent, authorization and grounded-feedback controls are cut to recover schedule	Low / High	Do not cut these controls; in PA3 the start stays disabled for unsupported media	Team	Any proposal to skip a consent test

Cut optional work (rankings, video, extra panels, FG-02 and stretch spikes) before cutting a Must requirement. If a Must requirement is at risk, tell the TA instead of relabeling a partial feature complete.

3.3 Lessons incorporated

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 From the PA1 lessons and the PA3 Sprint Review: agree interface contracts before parallel work; integrate and test small increments through the sprint; update Jira when work changes; plan demo preparation before the last days; do not put documents ahead of the graded slice. These are lessons recorded in the repository, not a new retrospective.

4. Project Plan

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14

4.1 Scrum process and communication

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 Scrum with sprint planning, at least two documented check-ins, a Sprint Review and a Retrospective. Messenger carries daily coordination, Google Meet carries meetings, GitHub PRs carry technical review, and Jira carries assignment and progress. Response times follow `docs/management/TEAM_CONTRACT.md`. Unfinished work is re-planned and reported, not marked Done.

Meeting evidence for PA3: weekly reports for 05/11/2026 and 10/11/2026 (`docs/pa3/WEEKLY_REPORT.md`) and the notes in `docs/pa3/MEETINGS/`. `SPRINT_PLANNING.md` in that folder currently describes PA4, and `SPRINT_REVIEW.md` is dated 2026-11-10 and awaits its 2026-11-13 update. These need correction before submission.

4.2 Five-sprint schedule

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 From `docs/ROADMAP.md`. Weeks are relative roadmap labels.

Sprint / PA	Weeks	Goal	Verification milestone
1 / PA1	1–2	Proposal, survey, contract, architecture, scaffold, CI	Done
2 / PA2	3–4	Plan, vision, Spec Kit init, AI and weekly report	Documents submitted; no code build
3 / PA3	5–6	Use-case model and specifications, prototypes, Build 1 then Build 2 (FG-03), `Changes.md`	Build 1 plus Build 2 with narrated video; documents done, FG-03 not met
4 / PA4	7–8	C4 and deployment, live audio and streaming STT	Build 3 (planned)
5 / PA5	9–10	Analysis, reports, testing, hardening, final demo	Build 4 and 15-minute live demo (planned)

4.3 Sprint 3 (PA3) schedule

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 Real dates from `docs/pa3/README.md` and `docs/pa3/CHECKLIST.md`.

Date	Event	Outcome
2026-11-03	Scope shrunk to FG-03; database first	Recorded in `Changes.md` rows 1.1 and 1.4
2026-11-04	Database first: migrations and seed target (SCRUM-41); Docker database (SCRUM-42)	SCRUM-42 done (PR #15); SCRUM-41 not on `pa3`
2026-11-05	Weekly report and first check-in; auth API and frontend shell target	Targets missed
2026-11-06	Room API and lobby UI target; CI migrate and seed job (SCRUM-43)	Job merged (PR #17), unverified
2026-11-10	Second check-in; FG-03 backend, frontend and E2E target; first Sprint Review status	FG-03 not done; first Sprint Review overstated progress and was corrected
2026-11-11	Revised plan target (SCRUM-35) and Lộc's review target	Plan was not revised; reassigned to Tuấn on 2026-11-14
2026-11-12	Video (SCRUM-39), use-case specs and PDFs targets	Specs done; video uploaded later (PR #47, 2026-11-14), content not checked; PDFs partial
2026-11-13	**Sprint 3 deadline (Friday)**	**Passed; Part E not met**
2026-11-14 to 11-15	Submission buffer for the zip and upload	Open now (this document is dated 11-14)

The deadline has passed. As of 2026-11-14 the following are still open: FG-03 implementation (E1; Tài's Java auth backend is merged via PR #53 but has no build file or tests and is unverified; no room API), the FG-03 Spec Kit spec (E2; `specs/001-frontend-baseline/` is merged via PR #45), the team check of the uploaded video (E3, SCRUM-39; https://youtu.be/wJxqi3J0phE, PR #47, visibility and content not checked), the E2E test (E4, SCRUM-38), per-member Spec Kit entries, a corrected weekly report, the tasks and prompts in the AI log (tools are recorded per member, PR #41), the meeting PDFs, rebuilt PDFs of the plan and vision, reviewer sign-offs for Parts C and D, the zip, the Git log export and the `pa3` to `dev` merge. Whether the buffer can cover them is not verified; what can be finished before the actual cutoff is an open decision (section 4.10).

4.4 Build 1 and Build 2 scope for FG-03

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 Build 1 was shrunk to what FG-03 needs. Source: `docs/pa3/Changes.md` section 1 (rows 1.1, 1.4, 1.5).

Kept in Build 1	Cut from Build 1
Migrations and seed with mock data (5 users, 10 topics, 1 room)	Refresh-token rotation
Register and login (Argon2id, access JWT)	Profile editing, CEFR screens
Create room, join by invite code, `LOBBY` to `READY`, 409 on a third user	`LIVE` and `ENDED` beyond the FG-03 start
Lobby UI	Separate Milestone 1 demo and smoke test (folded into the FG-03 video and E2E)

Build 1 Jira keys: SCRUM-24 auth, SCRUM-25 frontend, SCRUM-26 room API, SCRUM-27 lobby UI, SCRUM-28 state machine; SCRUM-29 and SCRUM-30 folded into FG-03; new SCRUM-41 database and seed, SCRUM-42 Docker database.

Build 2 is FG-03 end to end: topic selection, ready screen and a synchronized debate start, with UI to API to database connected, a Spec Kit trail and a narrated video. The start must not begin audio or transcription, because those are Build 3 and need dual consent (FR-13). The exact boundary of the start action is open decision D3. On 2026-11-14 Build 2 moved to PA4; PA3 Part E is FG-01 instead (Changes.md row 1.12).

Extra preparation (stretch, cut first): FG-02 topic API (SCRUM-44), LiveKit token spike (SCRUM-45), `STTAdapter` (SCRUM-46), `LLMAdapter` and grounding guard (SCRUM-47), microphone setup spike (SCRUM-48), CI migrate and seed (SCRUM-43).

Relevant requirement ids: FR-01 (register and authenticate), FR-10 (create room), FR-11 (join, capacity two), FR-12 (state graph and server-side validation); FR-13 to FR-15 belong to the later session part of FG-03. Contracts for Build 1 are in `docs/architecture/G1_SPEC_BUILD1.md`.

4.5 Work breakdown and status on 2026-11-14

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 "Not verified" means the repository sources disagree or give no evidence.

Jira	Work	Owner	Reviewer	Status
SCRUM-41	Alembic migrations and seed	Tài	Khoa	Open; not on `pa3`
SCRUM-42	Docker database	Lộc	not verified	Done (PR #15 merged)
SCRUM-43	CI migrate and seed	Lộc	not verified	Merged (PR #17); skips until SCRUM-41; unverified

Jira	Work	Owner	Reviewer	Status
SCRUM-24	Auth API	Tài	not verified	Java auth code (commit 3f585e0) merged into `pa3` (PR #53, `api/app/auth/`); no build file (no pom.xml or build.gradle) and no tests, so not verified to compile or run. Frontend auth (SCRUM-25) is on `pa3`, with browser tests not run on the build machine. Account recovery is gated off (PR #45). Java backend decision by Tài, 2026-11-14, confirmed 2026-11-14; backend specs owned by Tài (PR #43); AGENTS.md still says FastAPI
SCRUM-25	Frontend shell, register and login pages	Triết	not verified	Lobby work from `feature/SCRUM-25-ui-complete` merged as PR #28 (SCRUM-52); account UI merged as PR #39 (SCRUM-49); entry pages merged as PR #44 (SCRUM-50). Account recovery is gated off (PR #45). The shell and the register/login flow against a real auth API are not verified
SCRUM-26	Room API	Tài	not verified	Open; not on `pa3`
SCRUM-27	Lobby UI	Triết	not verified	Lobby UI merged as PR #28 (SCRUM-52); SCRUM-27 mapping verified by the team on 2026-11-14 and not connected to a room API
SCRUM-28	Room state machine and transition tests	Tuấn	Tài	Sprint Review says merged with unit tests; Tuấn's TODO says local, Tài review pending. Not verified

Jira	Work	Owner	Reviewer	Status
SCRUM-38	FG-03 backend, frontend and E2E	Tài, Triết; Khoa (E2E)	Lộc	Moved to PA4; Part E delivered as FG-01 instead
SCRUM-39	Narrated FG-03 video	Lộc	Khoa	Video uploaded (https://youtu.be/wJxqi3J0phE, PR #47); visibility and content verified by the team on 2026-11-14; FG-03 itself does not run on `pa3`
SCRUM-35	Revised plan and `Changes.md`	Tuấn (from Tài)	Lộc	Revised 0.2; reviewed by Lộc, 2026-11-14
SCRUM-35	Revised Vision (0.2)	Khoa	Lộc	PR #27 merged; final touches on `feature/SCRUM-35-vision-final`; sign-off pending
SCRUM-36	Use-case model	Tuấn	pending	Done (PR #19); sign-off pending
SCRUM-37	Use-case specifications and prototypes	Triết, Tuấn	pending	Done (PR #19); sign-off pending
SCRUM-40	AI usage log, weekly report, meetings, screenshots	Khoa, Lộc	Lộc	Partial
SCRUM-47	`LLMAdapter` and grounding guard	Tuấn	Khoa	Merged with unit tests per the Sprint Review; peer review not verified
SCRUM-44, 45, 46, 48	Extras: topic API, LiveKit token, `STTAdapter`, mic spike	Tài, Tài, Khoa, Triết	not verified	Stretch; no evidence on `pa3`
SCRUM-49 to 52	UI refinement packages (Tuấn, Tài, Khoa, Lộc)	as named	as named	SCRUM-49 account UI merged (PR #39, Tuấn); SCRUM-50 entry pages extracted and dead code removed (PR #44, Tài), partly done: shell and recovery pages not done; SCRUM-51 narrow preview summary and SCRUM-49 type/spacing roles merged (PR #46); SCRUM-52 lobby merged (PR #28); browser tests not run on the build machine (Chromium does not launch)

Jira	Work	Owner	Reviewer	Status
none	PA3-Group[GroupId] zip, Git log, pa3 to dev merge	Lộc	n/a	Missing; local PA3-Group2/ is partial

SCRUM-49 to 52 are UI refinement work. The graded Part E path is now FG-01 (specs/002-fg01-accounts).

4.6 Sprint 4 and Sprint 5 outline

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 Planned from docs/ROADMAP.md. Planned, not committed: detailed tasks, owners and dates are fixed at each sprint's planning meeting.

Sprint	Planned scope
Sprint 4 (PA4)	Revised use-case specifications with Changes.md; C4 context, container, component and deployment diagrams; LiveKit Cloud audio; Silero VAD and streaming STT adapter; Build 3 (FG-04 and FG-05) with a narrated demo. Any FG-03 gap carried over is fixed first
Sprint 5 (PA5)	Language Coach, Argument Coach, fact-checking adapter, consensus engine and grounding guards; reports and history; Playwright E2E, audio chaos tests and golden-set evaluation; Build 4 and the 15-minute live demo of all ten FGs; test plan and reflective report

The Sprint Review actions for PA4 apply: agree audio and STT interfaces before implementation, integrate regularly and record results, keep documents aligned with delivered work, and reserve time for demo and evidence.

4.7 Quality and verification approach

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 - A feature counts as done only when merged into pa3 through a reviewed PR and shown working. No claim without evidence: a PR number, test output, a screenshot or a video. - Tests: pytest for the state machine and room rules; PostgreSQL integration tests for room capacity and transitions; Playwright for the FG-03 browser journey (register, create, join, READY, third user gets 409); web checks from web/: npm run build, npm run lint, npm run test:e2e. - CI is the GitHub Actions workflow in .github/workflows/ci.yml. The migrate and seed job is unverified until SCRUM-41 lands (R-06). - Grounding (NFR-13): quote matching alone does not prove feedback is correct; human review is needed in Build 4. - Reviews: a merged PR counts as reviewed. The plan and vision sign-offs by Lộc are recorded on 2026-11-14; other reviewer sign-offs are not claimed until recorded. - No application test or benchmark result is reported in this plan. None was run while writing it.

4.8 Jira, source control and progress evidence

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 Jira project SCRUM is the task system. Branches are feature/SCRUM-<nn>-<short-name> (bug fixes bugfix/...) from pa3. PRs go into pa3, never into dev or main, and each needs a reviewer. Lộc merges pa3 into dev once, at the end of the sprint. No one commits code directly to pa3. Jira status and PR state are different observations and are reconciled together (R-03). The Jira screenshots are docs/pa3/jira-1.png and jira-2.png; live Jira was not read for this version.

4.9 Submission package and checklist

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 The shared status list is `docs/pa3/CHECKLIST.md`. The package `PA3-Group[GroupId].zip` must hold:

#	Item	Owner	Status
S1	`PA3-Group[GroupId]/` folder with all documents as `.md` and `.pdf`	Lộc	Missing; local folder partial and git-excluded
S2	Git log export	Lộc	Missing
S3	Source code without `node_modules`, build output or `.env`	Lộc	Code incomplete (FG-03)
S4	Spec Kit artifacts (`specs/`, plans, tasks)	all	Partial: `specs/001-frontend-baseline/` merged (PR #45); no FG-03 spec; per-member entries missing
S5	YouTube link for the Part E (FG-01) video	Lộc	Link exists (https://youtu.be/wJxqi3J0phE, PR #47); visibility and content not checked
S6	Zip of the package	Lộc	Missing
S7	`pa3` to `dev` merge after the package is complete	Lộc	Pending

PDFs come from `python report/pdf/build.py <file.md>` (Pandoc, Mermaid, `latexmk`). The rebuild of this plan's PDF is a separate step after this draft is reviewed.

4.10 Open decisions for the team

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14

Item	Decision needed	Follow-up
D1 — Remaining time	Which Part E items can be finished before the real cutoff, and whether to ask the TA about the missed 2026-11-13 deadline	Team, today
D2 — Cutoff	Exact Moodle date, time and timezone; official Group ID	Lộc
D3 — Build 2 start boundary	What the FG-03 "synchronized debate start" does without audio, keeping unsupported starts blocked	Tài, Triết, Lộc
D4 — Owner dependency	Who takes part of the SCRUM-41, 24, 26, 38 chain from Tài	Team
D5 — Database contract	The database design and G1 use different names (for example `full_name` and `target_cefr` against `display_name` and `cefr_level`). Agree a mapping before migrations are accepted	Tài, Tuấn, Lộc
D6 — Spec Kit entries	Each member's own Spec Kit entry, with real activity and date	Each member

Item	Decision needed	Follow-up
D7 — Status reconciliation	SCRUM-28 and SCRUM-47 merge and review state; Jira against the repository	Lộc, Tuấn

5. References and evidence baseline

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 Paths are repository-relative. No application tests were run and live Jira was not read when writing this version.

Ref.	Source	Use
S1	`docs/assignments/PA3_OVERVIEW.md`, `docs/pa3/README.md`	PA3 parts, points, rules, scope
S2	`docs/pa3/Changes.md`	Change rows 1.1 to 1.7 and Build 1 tables
S3	`docs/pa3/CHECKLIST.md`	Part-by-part status on 2026-11-14
S4	`docs/pa3/MEETINGS/SPRINT_REVIEW.md`, `docs/pa3/WEEKLY_REPORT.md`, `docs/pa3/AI_USAGE_LOG.md`	Retrospective, risks, evidence gaps
S5	`docs/pa3/<member>/TODO.md` (loc, triet, tai, khoa, tuan)	Owners, tasks, dates, statuses
S6	`docs/ROADMAP.md`, `docs/management/SPRINT_TASKS.md`, `docs/management/TEAM_CONTRACT.md`	Sprint plan, work breakdown, process
S7	`docs/requirements/VISION_DOCUMENT.md` (0.2)	Product scope
S8	`docs/architecture/SOFTWARE_ARCHITECTURE_SPEC.md`, `docs/architecture/G1_SPEC_BUILD1.md`	FR and NFR ids, Build 1 contracts
S9	`AGENTS.md`	Official roles and engineering rules

6. AI assistance declaration

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14 This revision was drafted with AI assistance from the repository sources listed in section 5, at the request of the lead while the owner was not present. The owner has not yet read or verified it. The owner must check every status and date against the repository, and record the AI use in `docs/pa3/AI_USAGE_LOG.md`. No Jira, commit, push or PR action was performed as part of this writing task.

7. Revision record

Performed: Trần Lê Anh Tuấn (24125107); Review: Nguyễn Đình Thiên Lộc (24125093), 2026-11-14; Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14

Version	Date	Change
0.1	2026-10-29	PA2 plan (SCRUM-20)
0.2	2026-11-14	PA3 revision (SCRUM-35), below

Changes in 0.2 against 0.1, with the matching `docs/pa3/Changes.md` rows:

#	Change in 0.2	Reason	Changes.md row
1	Owner changed to Tuấn, reviewer Lộc; PA2 performed and reviewed lines replaced by pending lines	Reassigned from Tài on 2026-11-14; no performance or review is claimed	Section 1 header
2	Build 1 moved to Sprint 3 and cut to what FG-03 needs (section 4.4)	PA2 had no code increment; PA3 grades one FG end to end	1.1
3	Spec Kit location at the repository root, no `src/` copy	One copy avoids drift	1.2
4	Real Sprint 3 dates, the passed deadline and open items (section 4.3)	Replaces the PA2 relative-week targets	1.3
5	Database first (SCRUM-41, 42) added to the plan	Every member runs the same data	1.4
6	Extras (SCRUM-43 to 48) marked as stretch, cut first	Reduce risk for Builds 3 and 4	1.5
7	Branch and PR rules (section 4.8)	Keep `dev` releasable	1.6
8	No TA feedback entries added	None received as of 2026-11-10	1.7
9	New: rubric mapping (2.3), risk register with the FG-03 gap, single owner and false status claims (3.2), work breakdown (4.5), quality approach (4.7), submission list (4.9), open decisions (4.10)	Brutal honesty on actual PA3 state	Not yet in `Changes.md`; add a row
10	PA2-only tables removed: Sprint 2 task close-out, proposed PA2 subtasks, PA2 review and packaging gates	Closed sprint; recoverable from Git history at `f789f5c`	3 (repository layout)